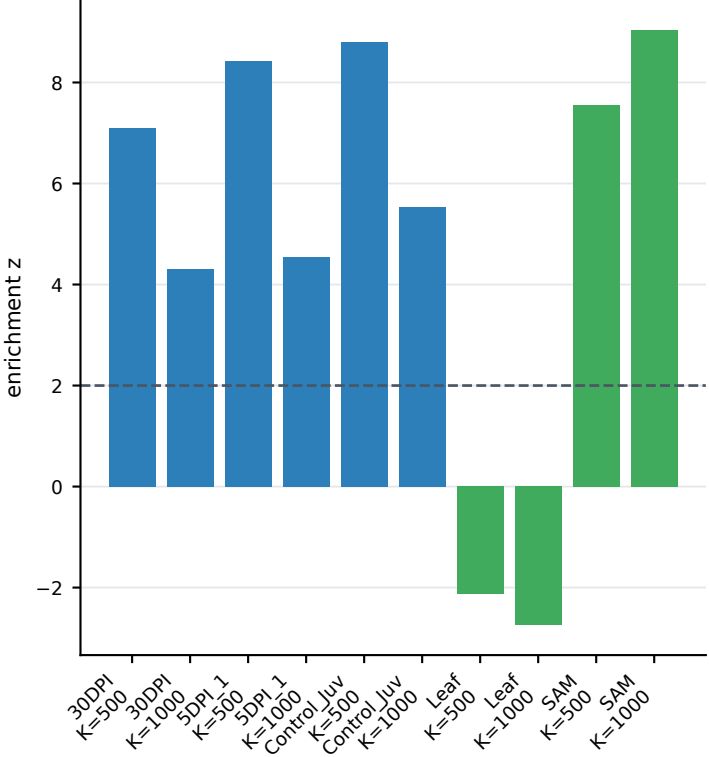


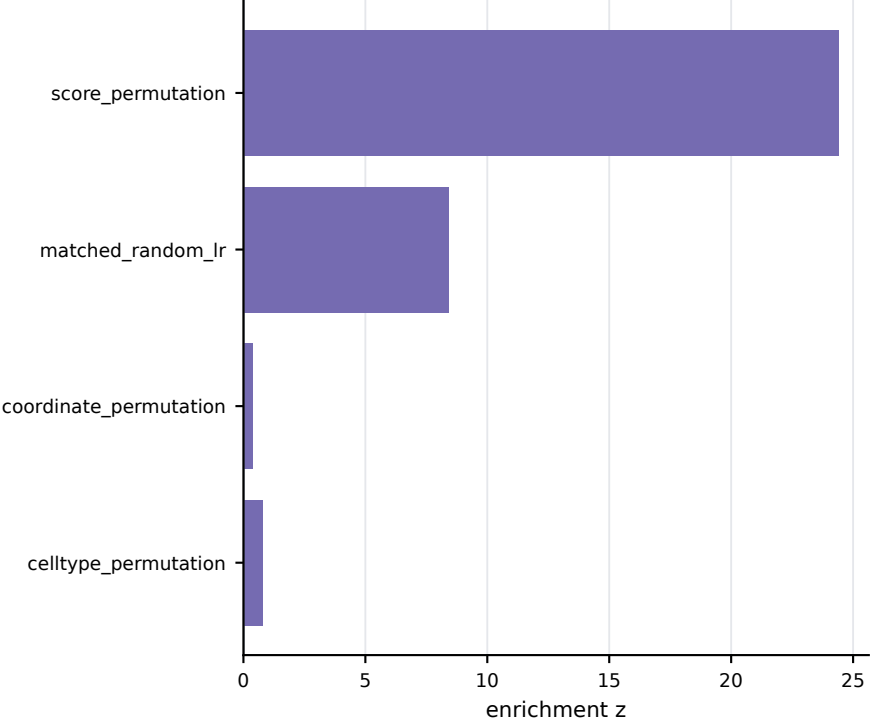
A Spatial validation datasets

source_dataset	section_id	n_cells_or_bins	n_groups	n_expression_genes
artista_axolotl	Stage44	1477	9	13098
artista_axolotl	Control_Juv	11698	17	19081
artista_axolotl	Adult	8243	16	22144
artista_axolotl	5DPI_1	8106	19	28741
artista_axolotl	30DPI	9252	17	19909
artista_axolotl	60DPI	10964	16	20038
sota_soybean	SAM	2239	5	13733
sota_soybean	Leaf	2391	5	25913

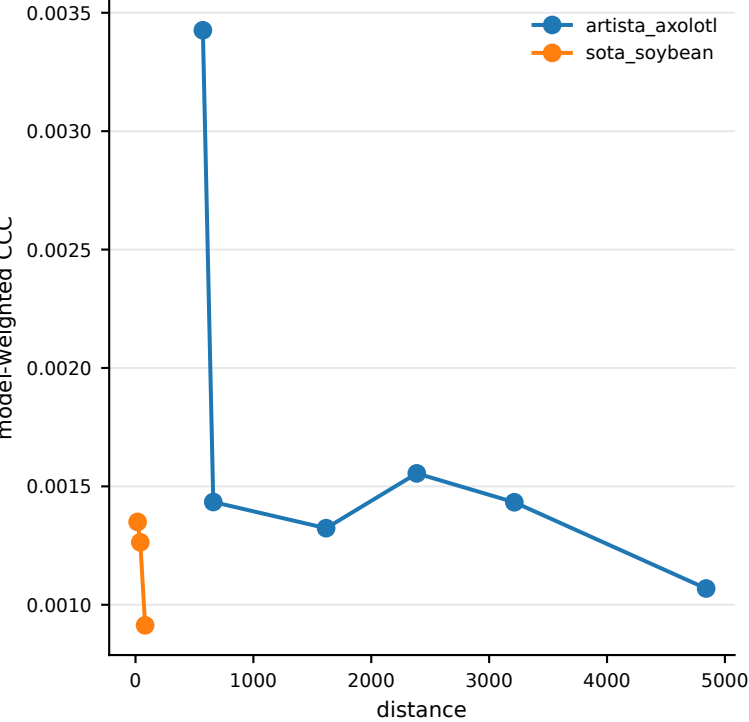
B Top-K spatial enrichment



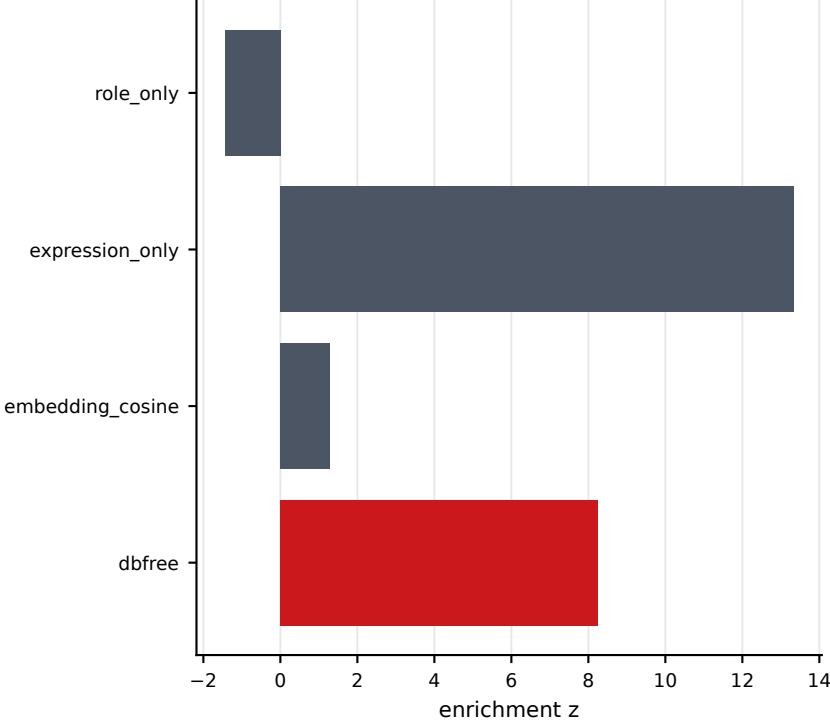
C Null model comparison



D Distance-decay curves



E Baseline ranking comparison



F Example spatial CCC validation

